Data File: 'processed1_truncated_data_with_rainfall_jenks.csv'

This is the input data file for this repository, and the point at which the code pipeline begins. It is a fully processed dataset. Every causative factor has already been extracted, discretized, and merged into a single table, one row per sample point (raster pixel location).

This is a truncated subset (~200,000 rows) of the full study dataset, provided for code review and reproducibility of the pipeline logic (class-imbalance handling, model training, evaluation). The full dataset used to generate the results reported in the manuscript contains **1,218,552 total samples**, of which **5,928 are landslide pixels** and **1,212,624 are non-landslide pixels** (~205:1 class imbalance). Because this is a subset, the class distribution and any metrics reproduced from this file will not exactly match the published results.

Run df["Landslide"].value_counts() on this file to see its actual class breakdown.

'Data.py' and 'avg_rainfall.py' both take this CSV directly as input and carry out class-imbalance handling. 'Undersampling-algos.py' and 'Oversampling-algos.py' further carry out model training/tuning, evaluation, and susceptibility-map generation. 'plots.py,' 'vulnerability.py,' and 'riskmap.py' consume outputs generated further downstream.

Column reference

Column	Description	Encoding
Unnamed: 0	Pandas row index carried over from a prior export step	Integer, no analytical meaning — safe to drop (scripts do this automatically)
Latitude	**Despite the name, this is the UTM Easting (X) coordinate**, not geographic latitude	Meters, EPSG:32644 (UTM Zone 44N)
Longitude	**Despite the name, this is the UTM Northing (Y) coordinate**, not geographic longitude	Meters, EPSG:32644 (UTM Zone 44N)
Lithology	Rock type at the sample location (from Geological Survey of India map)	1 = Phyllite, 2 = Sandstone with shale, 3 = Sandstone, 4 = Unconsolidated sediments, 5 = Shale with limestone, 6 = Granite, 7 = Quartzite, 8 = Quartzite alternating with shale, 9 = Dolomite, 10 = Limestone
Land_Cover	Land cover class at the sample location	1 = Cropland/Agricultural land, 2 = Waterbody, 3 = Settlement, 4 = Barren land, 5 = High/Dense vegetation (forest), 6 = Moderate vegetation (grassland), 7 = Low vegetation (grassland)
Plan_Curvature	Plan curvature of the terrain (DEM derivative)	Ordinal class, 6-class natural breaks: 1 = most concave, 6 = most convex

Profile_Curvature	Profile curvature of the terrain (DEM derivative)	Ordinal class, 6-class natural breaks: 1 = most concave, 6 = most convex
Aspect	Slope aspect (compass direction the terrain faces)	1 = Flat, 2 = North, 3 = Northeast, 4 = East, 5 = Southeast, 6 = South, 7 = Southwest, 8 = West, 9 = Northwest
Fault	Euclidean distance-buffer band from the nearest geological fault line	Ordinal class, 6 bands at 50 m intervals: 1 (≤50 m) → 6 (>250 m, up to 300 m band)
River	Euclidean distance-buffer band from the nearest river	Ordinal class, 6 bands at 40 m intervals: 1 (≤40 m) → 6 (>200 m, up to 240 m band)
Road	Euclidean distance-buffer band from the nearest road	Ordinal class, 6 bands at 40 m intervals: 1 (≤40 m) → 6 (>200 m, up to 240 m band)
Dem	Elevation (from Digital Elevation Model)	Ordinal class, 6-class natural breaks: 1 (≤887 m) → 6 (>3457 m)
Slope	Terrain slope angle (degrees)	Ordinal class, 6-class natural breaks: 1 (≤10.9°) → 6 (>47.3°)
TWI	Topographic Wetness Index	Ordinal class, 6-class natural breaks: 1 (≤3.38) → 6 (>13.65)
STI	Sediment Transport Index	Ordinal class, 6-class natural breaks: 1 (≤0) → 6 (>4.51)
SPI	Stream Power Index	Ordinal class, 6-class natural breaks: 1 (≤0) → 6 (>31.37)
Slope_Length	Slope length (meters)	Ordinal class, 6-class natural breaks: 1 (≤0) → 6 (>345.67 m)
Landslide	Ground-truth label — landslide occurrence at this point	1 = No landslide, 2 = Landslide
Rainfall_Jenks	30-year rainfall factor (maximum or average daily rainfall, depending on which pipeline generated the file. See Data.py vs avg_rainfall.py), classified via jenkspy	Ordinal class 1–6, Jenks natural breaks (thresholds computed dynamically from the rainfall raster, not fixed)

Notes

- For all "natural breaks" columns above, class **1 = lowest value range**, class **6 = highest value range**, of the underlying continuous variable at that location. Direction of association with landslide risk varies by factor (see manuscript SHAP analysis for feature-level interpretation).
- Latitude/Longitude are retained under those column names purely for compatibility with earlier processing steps; treat them as **projected UTM 44N coordinates**, not WGS84 lat/lon.